(Making it consistent with *general* relativity will prove much harder, and is the main subject of this book). In doing this, Dirac discovers an ulterior, profound simplification of our description of nature: the convergence between the notion of particles used by Newton and the notion of fields introduced by Faraday.

The cloud of probability which accompanies electrons between one interaction and another does resemble a field. Faraday and Maxwell's fields, in turn, are made up of grains: photons. Not only are the particles in a certain sense diffused in space like fields but the fields interact like particles. The notions of fields and particles, separated by Faraday and Maxwell, end up merging in quantum mechanics.

The way this happens in the theory is elegant: the equation of Dirac determines the values a variable can take. Applied to the *energy* of Faraday's lines, they tell us that this energy can only take on certain values, and not others. Since the energy of the electromagnetic field can take on only certain values, the field behaves like a set of packets of energy. These are precisely the quanta of energy introduced by Planck and Einstein thirty years earlier. The circle closes, and the story is complete. The equations of the theory, written by Dirac, account for the granular nature of light, which Planck and Einstein had intuited.

The electromagnetic waves are vibrations of Faraday's lines, but also, at a small scale, swarms of photons. When they interact with something else, as in the photoelectric effect, they manifest themselves as particles: to our eye, light rains in separate droplets, in single photons. Photons are the quanta of the electromagnetic field.

But the electrons and all the other particles of which the world is made are equally quanta of a field – a 'quantum field' similar to Faraday and Maxwell's, subject to granularity and to quantum probability. Dirac writes the equations for the field of the electrons and of the other elementary particles.[fn23] The sharp distinction between fields and particles introduced by Faraday vanishes.

The general form of quantum theory compatible with special relativity is thus called quantum field theory, and it forms the basis of today's particle physics. Particles are quanta of a field, just as photons are quanta of light. All fields display a granular structure in their interactions.

During the course of the twentieth century the list of fundamental fields was repeatedly updated, and today we have a theory called the *standard model* of elementary particles which describes almost all we see, with the exception of gravity,[fn24] in the context of quantum field theory. The development of this model occupied physicists for a good part of the last century, and represents in itself a wonderful adventure of discovery. I don't